

Moment-Compatible Readouts and Exceptional Pullbacks for the Bicomplex Signal Surface

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Abstract

Early drafts asserted that a power constraint and the moment assignment $(|z_1|^2, |z_2|^2) = (\cos t, 1 - \cos t)$, followed by a “geometric projection” $\mathbb{R}^4 \rightarrow \mathbb{R}^3$, produce the ruled signal surface. We prove that no readout which is real-analytic along the circle $z_1 = 0$ realizes the surface under that assignment, whatever state family is used. This excludes affine projections, every real polynomial readout and every polynomial in a bicomplex variable and its conjugates. The obstruction is sharp. An explicit continuous state family and an explicit C^∞ readout do realize the surface, and on either half of the source a real-analytic readout exists although no affine one does. Phase-invariant and mixed-state readouts fail without any regularity assumption. We also show that the exceptional pullback of every continuous map from a convex parameter domain to the plane is the extension by zero of the constant sheaf on its interior. The draft formula $i^*\Psi^!\mathbb{Z} \simeq \mathbb{Z}[-1]$ on the fold locus therefore holds at no point, and $\Psi^!\mathbb{Z}$ cannot detect folds. The fold is detected instead by failure of the comparison map required for cohomological smoothness. Finally, the log-scale versus linear-scale decorrelation gap is a ratio of normalizing window masses and involves no arithmetic.

1 Sources and scope

Three historical records are kept distinct; their SHA-256 values are

v12 PDF, 43 pp. (authority)	4ebb8f0a7d6e88a6c5cd97f65d224b4e938867cabad54f057d3ceafb24167d5a
v7 PDF, 26 pp.	6cefa5258257f3a23e35b5dd5cbe123b7af0ed3319c3a614b1c8d54c94bd654f
ancestor TeX	4c0a5b5a8287ab5559c892b2ee21c170984e5f9649481018f851095118a1f016

The ancestor TeX has 1,908 lines. The v7 draft, dated July 2026 on its title page, is registered here as a non-authoritative historical draft. It is neither v12 nor a demonstrated product of the ancestor TeX. Equal numbering does not imply equal content. Page numbers below are physical pages of the cited PDF.

All three records contain the moment assignment of §2: v7 §3 and Definition 3.1 on p. 2, the ancestor §3, and v12 §3 and Definition 3.1 on p. 4. The exceptional-pullback, monodromy-
endomorphism and log-window statements of §§3–4 are v7 Proposition 8.3 and Remarks 8.4 and 8.16 (pp. 10–11, 18), Definition 8.9 to Corollary 8.11 (pp. 14–15) and Proposition 8.13 with Remark 8.14 (pp. 15–17). The ancestor and v12 no longer assert the exceptional-pullback identity; they do not decide it either. The results below are new constructions, proofs and counterexamples. They are not recoveries of missing historical derivations.

Throughout,

$$S(t, u) = (c + 2u(1 - c), (1 - u)s, uQ), \quad c = \cos t, \quad s = \sin t, \quad Q = \sqrt{c(2 - c)},$$

on $D = [-\pi/2, \pi/2] \times [0, 1]$. Put $D_\circ = (-\pi/2, \pi/2) \times [0, 1]$ and $D_+ = [0, \pi/2] \times [0, 1]$. The edges and the polar ruling map to

$$S(\pm \frac{\pi}{2}, u) = (2u, \pm(1 - u), 0), \quad S(0, u) = (1, 0, u). \quad (1)$$

The two edge segments lie on distinct lines through the apex $(2, 0, 0)$. The ruling line $\{(1, 0, z)\}$ is skew to both edge lines.

2 The ancestral moment assignment

Let $S^3 = \{z \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}$, and let $C_0 = \{(0, w) : |w| = 1\}$ and $C_1 = \{(w, 0) : |w| = 1\}$. The bicomplex idempotent coordinates identify the algebra with $\mathbb{C} \oplus \mathbb{C}$; we write the two components as z_1, z_2 . A real polynomial in a bicomplex variable and its three standard conjugates is a real polynomial in $\Re z_j, \Im z_j$.

Definition 2.1. Let $D' \subset D$. A *moment-compatible state family* on D' is any map $z : D' \rightarrow S^3$ with $|z_1(t, u)|^2 = \cos t$. No continuity is assumed. A *readout* is a map $P : U \rightarrow \mathbb{R}^3$ on a set $U \subset \mathbb{C}^2$ containing $z(D')$; it *realizes* S if $P \circ z = S$ on D' . It is *real-analytic along* C_0 if $C_0 \subset U$ and $\beta \mapsto P(0, e^{i\beta})$ is real-analytic.

Every state with $t = \pm\pi/2$ lies on C_0 , and every state with $t = 0$ lies on C_1 . This alone already excludes the gauge-invariant readouts.

Proposition 2.2 (Phase-invariant and mixed-state readouts). *Suppose D' contains two points $(\pi/2, u_1) \neq (\pi/2, u_2)$, or two such points at $t = -\pi/2$. No readout with $P(e^{i\gamma}z) = P(z)$ realizes S on D' . In particular no function of the density matrix zz^* does, including all Hermitian expectation values. If density matrices $\rho(t, u) \geq 0$ of trace one with $\rho_{11} = \cos t$ replace pure states, no function of ρ realizes S .*

Proof. C_0 is a single orbit of the global phase, so such a readout is constant on C_0 . By (1), however, S takes distinct values at the two edge points. For mixed states, $\rho_{11} = 0$ and positivity force $\rho_{12} = 0$, so $\rho = \text{diag}(0, 1)$ at both points. $\square$

Lemma 2.3 (Analytic curves and lines). *Let $p : J \rightarrow \mathbb{R}^3$ be real-analytic on an open interval J or on the circle. For every line ℓ , either $p(J) \subset \ell$ or $p(J) \cap \ell$ is countable. Hence $p(J)$ cannot contain uncountable subsets of two distinct lines.*

Proof. Choose $x_0 \in \ell$ and unit normals n_1, n_2 to ℓ . If $p(J) \not\subset \ell$, some $f_k = n_k \cdot (p - x_0)$ is a nonzero real-analytic function, so its zero set is discrete in J and hence countable. It contains $p^{-1}(\ell)$. If $p(J)$ met two distinct lines in uncountable sets, it would lie in both lines. Their intersection has at most one point, a contradiction. $\square$

Theorem 2.4 (Analytic rigidity). *Let z be a moment-compatible state family and P a readout that realizes S on D' .*

1. *If $D' = D$, then P is not real-analytic along C_0 . More generally, P is not real-analytic on any connected open arc of C_0 containing all edge states $z(\pm\pi/2, u)$.*
2. *If $D' = D_\circ$, $C_0 \subset U$ and $P|_{U \cap S^3}$ is continuous at each point of C_0 , then P is not real-analytic along C_0 .*

Proof. (1) Each $z(\pm\pi/2, u)$ lies on C_0 . Hence $P(C_0)$, or the image of the arc, contains both edge segments in (1). They are uncountable subsets of distinct lines, which Lemma 2.3 forbids.

(2) Fix u and $\epsilon = \pm 1$, and take $t_n \rightarrow \epsilon\pi/2$ in $(-\pi/2, \pi/2)$. The states $z(t_n, u) \in S^3$ have a subsequence converging to some z^* . Since $|z_1|^2 = \cos t_n \rightarrow 0$, $z^* \in C_0$. Continuity at z^* gives $P(z^*) = \lim S(t_n, u) = S(\epsilon\pi/2, u)$. Thus both closed edge segments lie in $P(C_0)$, and (1) applies. $\square$

Corollary 2.5 (Excluded projections). *Under the ancestral assignment on D or $D_\circ$, S is not realized by any affine map $\mathbb{R}^4 \rightarrow \mathbb{R}^3$, any real polynomial map, any polynomial in a bicomplex variable and its conjugates, or any map real-analytic on a neighbourhood of C_0 , such as a rational map regular near C_0 . On D it also excludes stereographic projection from a point of C_0 that avoids the edge states. Consequently the unspecified “Geometric Projection” of v7 and v12 Definition 3.1 cannot be completed by any such map.*

The obstruction is analyticity, not topology or smoothness.

Theorem 2.6 (C^∞ realization). *Put $\theta = \pi/8$. Let $\Lambda, \psi, \varrho, \kappa : [0, 1] \rightarrow [0, 1]$ be smooth, with $\Lambda = 0$ on $[0, \frac{1}{4}]$, $\Lambda = 1$ on $[\frac{3}{4}, 1]$, Λ strictly increasing on $(\frac{1}{4}, \frac{3}{4})$, $\psi = 0$ on $[0, \frac{1}{3}]$, $\psi = 1$ on $[\frac{2}{3}, 1]$, $\varrho = 0$ on $[0, \frac{3}{4}]$, $\varrho = 1$ on $[\frac{7}{8}, 1]$, $\kappa = 1$ on $[0, \frac{1}{8}]$ and $\kappa = 0$ on $[\frac{1}{4}, 1]$. Let χ be a smooth 2π -periodic function equal to 1 on $[-\frac{\pi}{8}, \frac{3\pi}{8}]$ and to -1 on $[\frac{7\pi}{8}, \frac{11\pi}{8}]$. Let $\epsilon = 0$ for $t \geq 0$ and $\epsilon = 1$ for $t < 0$, and define*

$$z(t, u) = (\sqrt{c} e^{i\theta\Lambda(c)u}, \sqrt{1-c} e^{i(\epsilon\pi + \theta(1-\Lambda(c))(1+u))}).$$

This is a moment-compatible state family, continuous on D and smooth on $D_\circ$. It is also smooth up to $t = \pm\pi/2$ in the completed edge charts $r = Q$, since $\sqrt{C(r)} = r/\sqrt{1 + \sqrt{1-r^2}}$ there. With $c = |z_1|^2$, set

$$\begin{aligned} U_A &= \frac{\arg z_1}{\theta\Lambda(c)}, \quad U_B = \frac{\arg z_2^2}{2\theta(1-\Lambda(c))} - 1, \quad \mathcal{U} = \psi(c)U_A + (1-\psi(c))U_B, \\ \Sigma &= \varrho(c) \Re z_2 \sqrt{1+c} + (1-\varrho(c)) |z_2| \chi(\arg z_2) \sqrt{1+c}, \\ \mathcal{Q} &= \kappa(c) \Re z_1 \sqrt{2-c} + (1-\kappa(c)) \sqrt{c(2-c)}, \end{aligned}$$

using principal arguments. Each summand is understood to be zero where its cutoff vanishes. Then

$$P = (|z_1|^2 + 2\mathcal{U}|z_2|^2, (1-\mathcal{U})\Sigma, \mathcal{U}\mathcal{Q})$$

is smooth on an open neighbourhood of $z(D)$, and $P \circ z = S$ on all of D . Multiplying by a smooth bump equal to 1 near $z(D)$ gives a C^∞ readout on $\mathbb{C}^2$.

Proof. Continuity at $t = 0$ holds because $\Lambda = 1$ for $c \geq \frac{3}{4}$ makes $z_2 = \operatorname{sgn}(t)\sqrt{1-\cos t} = \sqrt{2}\sin(t/2)$, which is smooth. At $t = \pm\pi/2$ only $\sqrt{c} \rightarrow 0$ enters. On the state image, $\arg z_1 = \theta\Lambda u \in [0, \theta]$, $\arg z_2^2 = 2\theta(1-\Lambda)(1+u) \in [0, \pi/2]$, and $\arg z_2$ lies in $\epsilon\pi + [0, \pi/4]$. Hence $U_A = u$ wherever $\Lambda > 0$ and $U_B = u$ wherever $\Lambda < 1$. The cutoff ψ uses U_A only for $c \geq \frac{1}{3}$, where $\Lambda > 0$ and $z_1 \neq 0$. It uses U_B only for $c \leq \frac{2}{3}$, where $\Lambda < 1$ and $|z_2|^2 \geq \frac{1}{3}$. Thus $\mathcal{U}$ is smooth near the image and equals u .

Where $\varrho > 0$ we have $c \geq \frac{3}{4}$, so $\Lambda = 1$ and z_2 is real. Then $\Re z_2 \sqrt{1+c} = \operatorname{sgn}(t)\sqrt{1-c^2} = \sin t$. Where $\varrho < 1$, $|z_2|^2 \geq \frac{1}{8}$ and $\chi(\arg z_2) = \operatorname{sgn} t$, giving the same value. Where $\kappa > 0$, $c \leq \frac{1}{4}$, so $\Lambda = 0$, $z_1 = \sqrt{c}$ and $\Re z_1 \sqrt{2-c} = Q$. Each summand is smooth on the union of two open sets: one on which its decoder is smooth, and one on which its cutoff vanishes. The two definitions agree on the overlap, and the union contains the image. Substitution gives $P \circ z = S$. The image is compact, so a bump exists. $\square$

A finite replay on 801×81 source labels confirms the identity to 1.4×10^{-14} and the injectivity of the sampled labels. On C_0 this readout traces the two legs of the V on disjoint arcs, joined by the flat cutoff χ . That is exactly what Lemma 2.3 forbids for an analytic readout.

Proposition 2.7 (One half of the source). *(1) Let D' contain at least three points of the ruling $\{0\} \times [0, 1]$ and at least three points of one edge $\{\epsilon\pi/2\} \times [0, 1]$. Then no affine readout realizes S on D' under the ancestral assignment. This applies in particular to D_+ , which contains the single interior cross-cap P_+ .*

(2) On D_+ there are a real-analytic moment-compatible state family and a readout that is real-analytic on an open neighbourhood of its image and realizes S .

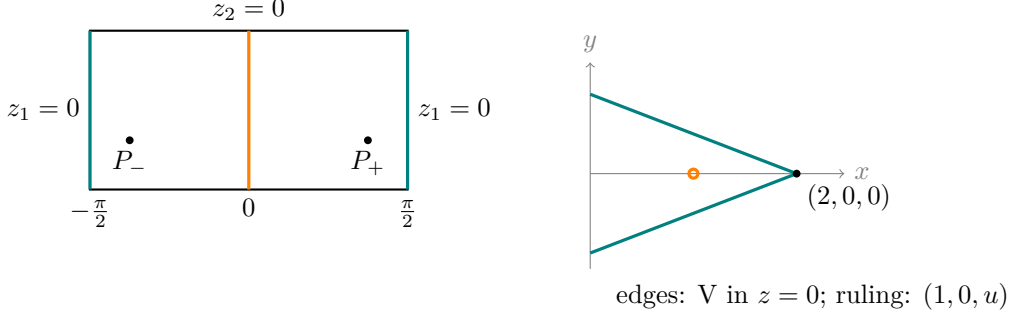


Figure 1: Ancestral powers vanish on the two edges ($z_1 = 0$) and on the polar ruling ($z_2 = 0$). The edges map to a V in the plane $z = 0$, which no analytic image of C_0 can contain. The ruling (orange, vertical over $(1, 0)$) maps to a segment on a line skew to both legs.

Proof. (1) Write $P(z) = b + L(z)$. On C_1 the readout traces $b + L(\cos \alpha, \sin \alpha, 0, 0)$, an ellipse centred at b , possibly degenerate. It contains three points of the ruling line. A nondegenerate ellipse meets a line in at most two points, so the ellipse is a segment centred at b on that line. Hence b lies on the ruling line. The same argument on C_0 puts b on the edge line. These lines are skew, a contradiction.

(2) Fix $0 < \theta < \pi$ and set $z = e^{i\theta u}v(t)$, where $v = (\sqrt{\cos t}, \sqrt{2}\sin(t/2))$. In the edge chart $r = Q$ it becomes $v = (r/\sqrt{1 + \sqrt{1 - r^2}}, (1 - r^2)^{1/4})$. Both expressions are real-analytic and extend across $t = 0$ and $r = 0$, and so does S in the completed charts. The map is an immersion, because $\partial_u z = i\theta z$ is a purely imaginary multiple of $e^{i\theta u}v$ while $\partial_t z = e^{i\theta u}v'$ has $v' \neq 0$ real. It is injective on D_+ : if $e^{i\theta u}v = e^{i\theta u'}v'$ with v, v' real and nonnegative, the phase quotient is positive, so $u = u'$ and then $t = t'$. An immersion injective on a compact set is an embedding of a neighbourhood. The nearest-point retraction to a real-analytic embedded surface is real-analytic on a tube, by the analytic inverse function theorem applied to $(x, n) \mapsto x + n$ on the normal bundle. Compose it with the inverse embedding and the extended map S . $\square$

Remark 2.8 (What extra data the historical model needs). Theorem 2.4 gives a necessary edge criterion for any power assignment $w = |z_1|^2$. Analytic readouts require $S(w^{-1}(0))$ to lie in the image of an analytic loop, and phase-invariant readouts require it to be a point. The same holds for $1 - w$. The 0.02 assignment $(1 - u, u)$ passes: its vanishing edges $u = 1$ and $u = 0$ map to arcs of the circles $(x - 1)^2 + z^2 = 1$ and $x^2 + y^2 = 1$, and its quadratic readout maps the vanishing circles onto these circles. The ancestral assignment fails, because $w^{-1}(0)$ consists of both edges.

Three additional inputs therefore remain, and none is supplied by the moment map. One may use a non-analytic switching readout as in Theorem 2.6. One may restrict to one half of the source, which loses a cross-cap and the reflection pairing. Or one may change the powers, as continuation 0.02 does. Approximation is cheap and carries no canonical content: by Weierstrass, real polynomial readouts satisfy $\sup_D |P_\varepsilon \circ z - S| < \varepsilon$ for every $\varepsilon > 0$ with the family of Theorem 2.6, yet none is exact. Whether a quadratic readout exists on D_+ is left open.

3 Exceptional pullback and folds of the observation map

We work in the six-functor formalism on finite-dimensional locally compact Hausdorff spaces with $D(X) = D(\text{Sh}(X; \mathbb{Z}))$ [1, Example 5.9]. Write $\omega_X = a_X^! \mathbb{Z}$ and $\mathbb{D}_X = \text{RHom}(-, \omega_X)$. We use three standard facts. First, Verdier duality gives $f^! \mathbb{D}_Y \simeq \mathbb{D}_X f^*$ [2, Prop. 4.6.9(3)], [3, Ch. III]. Second, the stalk of ω_X at x is local homology $H_*(X, X \setminus x)$, and $\omega_X \simeq \text{or}_X[n]$ for an n -manifold without boundary [2, Rem. 4.6.19, Cor. 4.6.20]. Third, $f^!$ is local on the source, and

open immersions satisfy $j^! = j^*$ and are cohomologically smooth [1, Rem. 5.4]. Cohomological smoothness of f requires that $f^! \mathbb{Z} \otimes f^*(-) \rightarrow f^!(-)$ be an isomorphism, that $f^! \mathbb{Z}$ be invertible, and that both survive base change [1, Def. 5.1]. The draft’s paraphrase “ $f^! \mathbb{Z} \simeq \mathbb{Z}[d]$ ” records only the second condition.

Theorem 3.1 ($f^! \mathbb{Z}$ sees only dimensions and boundary). (1) For any continuous $f : X \rightarrow Y$ between topological manifolds without boundary of dimensions n and m , $f^! \mathbb{Z}_Y \simeq \text{or}_X \otimes f^* \text{or}_Y[n - m]$.

(2) Let $D' \subset \mathbb{R}^2$ be a locally closed convex set with nonempty interior, for instance D or $D_\circ$, and let $j : \text{int } D' \rightarrow D'$ be the inclusion. For every continuous $\Psi : D' \rightarrow \mathbb{R}^2$,

$$\Psi^! \mathbb{Z}_{\mathbb{R}^2} \simeq j_! \mathbb{Z}_{\text{int } D'}.$$

Its stalk is $\mathbb{Z}$ in degree 0 at interior points and 0 at the remaining points.

Proof. (1) Since ω_Y is invertible, $\mathbb{Z}_Y \simeq \mathbb{D}_Y \omega_Y$. Hence

$$f^! \mathbb{Z}_Y \simeq \mathbb{D}_X f^* \omega_Y = \text{RHom}(f^* \text{or}_Y[m], \text{or}_X[n]),$$

and rank-one orientation systems are self-dual.

(2) For an interior point, $H_*(D', D' \setminus x) = \mathbb{Z}[2]$. For any other point x , a point strictly between an interior point and a point of D' is interior, so $D' \setminus x$ is star-shaped about an interior point and hence contractible. Local homology then vanishes. Thus $\omega_{D'}$ restricts to $\mathbb{Z}[2]$ on the oriented interior and has zero stalks elsewhere. The counit $j_! j^* \omega_{D'} \rightarrow \omega_{D'}$ is therefore a stalkwise isomorphism, and $\omega_{D'} \simeq j_! \mathbb{Z}_{\text{int } D'}[2]$. Now $\Psi^! \mathbb{Z} \simeq \mathbb{D}_{D'} \Psi^* \omega_{\mathbb{R}^2} = \text{RHom}(\mathbb{Z}_{D'}[2], \omega_{D'}) = \omega_{D'}[-2]$. $\square$

Corollary 3.2 (The draft shift). The observation map $\Psi = (N, M)$ of v7 Proposition 8.3 is defined and smooth on $D_\circ$; since $M = P/Q$ and $Q = 0$ at $t = \pm\pi/2$, it is not defined on all of D . For every inclusion $i : L \rightarrow D_\circ$, $i^* \Psi^! \mathbb{Z}$ is $\mathbb{Z}$ on $L \cap \text{int } D_\circ$ and 0 on the sides $u = 0, 1$. The claimed $i^* \Psi^! \mathbb{Z} \simeq \mathbb{Z}[-1]$ on the fold locus holds at no point. The “relative dimension” is 0 at every interior point, fold or not, contrary to v7 Remark 8.16. If $L \subset \text{int } D_\circ$ is an embedded open arc, then $i^! \Psi^! \mathbb{Z} \simeq \text{or}_L[-1]$. The shift $[-1]$ in the draft’s local-cohomology sketch is the codimension of L ; it holds equally for arcs of regular points.

Proof. Restrict Theorem 3.1(2) with $D' = D_\circ$. For the arc, $i^! \mathbb{Z}_{\text{int } D_\circ} = i^! \omega_{\text{int } D_\circ}[-2] = \omega_L[-2] = \text{or}_L[-1]$. $\square$

The interior critical locus of Ψ was determined in the published Stokes paper [4, Theorem 2.4]. Every point of the symmetry segment with $0 < u < 1$, and every interior point of the rational branch with $c_b < c < 1$, is an ordinary Whitney fold. At $(0, 0)$ the ordinary-fold condition fails. The verifier replays the Jacobian factorization and $N_{tt}(0, u) = -2u$.

Proposition 3.3 (Where the fold is visible). Let $x \in \text{int } D_\circ$ be an ordinary fold point of Ψ . Choose disc neighbourhoods $V \ni x$ and $W \ni \Psi(x)$ on which $f = \Psi|_V$ is homeomorphic to $(\xi, \eta) \mapsto (\xi, \eta^2)$. Let L be the fold arc and $W' \subset W$ the open side of the discriminant containing $f(V \setminus L)$, with inclusion j' . For $G = j'_! \mathbb{Z}_{W'}$:

1. $f^! \mathbb{Z}_W \simeq \mathbb{Z}_V$ is invertible and unshifted;
2. $f^* G \simeq (\mathbb{Z}_{V \setminus L})_!$ and $f^! G \simeq \mathbb{Z}_V$;
3. the comparison map $f^! \mathbb{Z}_W \otimes f^* G \rightarrow f^! G$ is injective with cokernel $i_* \mathbb{Z}_L$; its stalks at L are $0 \rightarrow \mathbb{Z}$.

Thus condition (1) of [1, Def. 5.1] fails at every ordinary fold point, while condition (2) holds. At regular interior points Ψ is a local homeomorphism and is cohomologically smooth with $\Psi^! \mathbb{Z} \simeq \mathbb{Z}$.

Proof. (1) is Theorem 3.1(1). Pullback commutes with extension by zero, and $f^{-1}(W') = V \setminus L$. Let $k : H \rightarrow W$ be the closed side $\overline{W'} \cap W$. Its local homology vanishes on its edge, so $\mathbb{D}_W(\mathbb{Z}_H) = k_*\omega_H \simeq G[2]$. Since $f(V) \subset H$,

$$f^!G[2] \simeq f^!\mathbb{D}_W(\mathbb{Z}_H) \simeq \mathbb{D}_V(f^*\mathbb{Z}_H) = \mathbb{D}_V\mathbb{Z}_V = \mathbb{Z}_V[2].$$

Off L , f is locally an open immersion, so the comparison map is an isomorphism there. A morphism $(\mathbb{Z}_{V \setminus L})! \rightarrow \mathbb{Z}_V$ is a locally constant function on $V \setminus L$. Being invertible there, it takes values ± 1 . Its image is $(\mathbb{Z}_{V \setminus L})!$, with cokernel $i_*\mathbb{Z}_L$. $\square$

The fold is also seen by $f_*\mathbb{Z}_V$, whose stalk ranks are 2, 1, 0 across the discriminant, and on the surface side by the normalization defect sheaf already proved in v12/v13. It is not seen by $f^!\mathbb{Z}$. Remark 8.4's appeal to Verdier duality for the zero count is also unnecessary. The total index of the interior zeros of $F = N + iM$ equals the winding number of F along any loop in $D_\circ$ that encloses $P_\pm$ and avoids $(0, 1)$. This is elementary degree theory.

Proposition 3.4 (Carriers for the ± 1 sectors). *(1) Let X be connected. Then $\text{End}_{D(X)}(\mathbb{Z}_X) = H^0(X; \mathbb{Z}) = \mathbb{Z}$, every involution of $\mathbb{Z}_X$ is $\pm \text{id}$, and $\mathbb{Z}_X$ has no splitting into two nonzero summands. Its characteristic polynomial is $t - 1$ or $t + 1$. For the constant condensed sheaf on the compact connected image, the degree-zero endomorphisms are again the continuous maps to $\mathbb{Z}$, so the conclusion is unchanged.*

(2) Let $X = D_\circ \setminus \{P_+, P_-, (0, 1)\}$, the set where $F \neq 0$, and let $\pi : \tilde{X} \rightarrow X$ be the double cover $\{\chi^2 = F\}$. Then $\pi_\mathbb{Z}$ is a rank-two local system with deck involution τ , and it is indecomposable over $\mathbb{Z}$. After inverting 2, $\pi_*\mathbb{Z}[\frac{1}{2}] \simeq \mathbb{Z}[\frac{1}{2}]_X \oplus \mathcal{L}[\frac{1}{2}]$, where $\mathcal{L}$ is the sign system with monodromy -1 around P_+ and around P_- . In particular $\mathcal{L}$ does not extend over either point.*

(3) A Whitney fold involution $(\xi, \eta) \mapsto (\xi, -\eta)$ fixes the fold arc pointwise, whereas τ has no fixed points. Neither induces the other.

Proof. (1) Hom between sheaves in degree zero is $\Gamma(X, \mathcal{H}om(\mathbb{Z}, \mathbb{Z}))$. For the condensed constant sheaf on a compact Hausdorff space, global sections are continuous maps to the discrete group $\mathbb{Z}$. Idempotents of $\mathbb{Z}$ are 0, 1. (2) $N = 0$ exactly on $c = 1$ and on $u = (1 - c)/2$. On the first, $M = 1 - u$; on the second, M is a positive multiple of $-(c^2 - 3c + 1)$. Hence the zeros of F in $D_\circ$ are $(0, 1)$ and $P_\pm$. The set $D_\circ \setminus \{(0, 1)\}$ is star-shaped about an interior point, so $\pi_1(X)$ is free on loops around $P_\pm$. The winding of F is ∓ 1 there (Lean `actual_dipole`; finite replay). Hence the monodromy of π is the surjection $\pi_1(X) \rightarrow C_2$, and the stalk of $\pi_*\mathbb{Z}$ is the regular module $\mathbb{Z}[C_2] = \mathbb{Z}[x]/(x^2 - 1)$. Its only idempotents are 0 and 1 (verifier), so it is indecomposable. Over $\mathbb{Z}[\frac{1}{2}]$ the projectors $(1 \pm x)/2$ split it. (3) The fixed-point sets differ. $\square$

Thus v7 Definition 8.9 to Corollary 8.11 place a rank-two ± 1 decomposition on the rank-one constant sheaf, where (1) forbids it. The correct carrier of a -1 sector is the sign system of the square-root cover, and it splits off only after inverting 2. The draft writes its projectors over $\mathbb{Z}[\frac{1}{2}]$ but its module over $\mathbb{Z}$. No map from the Mellin space $L^2([1, \infty))$ to either carrier is supplied, so the claimed inclusion of the Mellin domain in the $+1$ sector has no object to refer to.

4 Log-scale versus linear-scale windows

Let $K(\delta, L) = L^{-1} \int_0^L e^{-i\delta\tau} d\tau$ for real $\delta \neq 0$. Then $|K(\delta, L)| = |\sin(\delta L/2)|/(|\delta|L/2) \leq 2/(|\delta|L)$.

Proposition 4.1 (Window masses, not arithmetic). *Let $T > 1$.*

1. $\frac{1}{\log T} \int_1^T x^{-i\delta \frac{dx}{x}} = K(\delta, \log T)$ and $\frac{1}{T} \int_0^T e^{-i\delta t} dt = K(\delta, T)$. *The two v7 bounds are one bound, $2/(|\delta| \cdot \text{window mass})$, evaluated at masses $\log T$ and T . Their ratio $T/\log T$ is the same for every real δ , whether or not $\delta = \log(n/m)$. The exact ratio of the moduli is $(T/\log T) |\sin(\delta \log T/2)/\sin(\delta T/2)|$. Along $T_k = (2k + 1)\pi/|\delta|$ the second factor equals $|\sin(\delta \log T_k/2)|$, whose $\liminf$ is 0 and $\limsup$ is 1.*

2. For the same characters $x^{-i\delta}$ on the same interval $[1, T]$ with Lebesgue measure,

$$\frac{1}{T-1} \int_1^T x^{-i\delta} dx = \frac{T^{1-i\delta} - 1}{(1-i\delta)(T-1)},$$

whose modulus tends to $(1 + \delta^2)^{-1/2} \neq 0$. There is no decorrelation at all; with Haar measure the decay is $O(1/\log T)$. The ordering claimed by Remark 8.14 is reversed.

3. For the time phasors $e^{-i\delta t}$ on $[1, T]$ with Haar measure, $|\frac{1}{\log T} \int_1^T e^{-i\delta t} \frac{dt}{t}| \leq \frac{2}{|\delta| \log T}$.

Proof. (1) Substitute $x = e^\tau$. Along T_k , $\delta \log T_k/2$ tends to infinity with increments tending to zero, so it comes arbitrarily close to every residue modulo π . (2) Integrate directly. (3) Integrate by parts once; the boundary terms and the remaining integral contribute at most $2/|\delta|$ in total. $\square$

In no reading does integrality enter. Moreover $\mathbb{N}$ is not a subgroup of $(\mathbb{R}_{>0}, \times)$; the subgroup it generates is $\mathbb{Q}_{>0}$, which is dense. The proposed aliasing and Weil-group interpretation of Remark 8.14 therefore has no quantity to explain.

5 Status and open questions

Historical item	Result here	Evidence
Moment derivation of S (v7/ancestor/v12 §3, Def. 3.1)	Phase-invariant, mixed-state and analytic readouts excluded; C^∞ realization exists; one half admits analytic but not affine readouts	Written proofs; exact/finite replay
v7 Prop. 8.3, Rem. 8.16	$\Psi^! \mathbb{Z} \simeq j_! \mathbb{Z}_{\text{int } D'}$ for every continuous Ψ on a convex D' ; draft shift false; fold detected by the comparison map	Written proofs from cited duality theorems
v7 Def. 8.9–Cor. 8.11	$\text{End}(\mathbb{Z}_X) = \mathbb{Z}$; correct carrier is the indecomposable $\pi_* \mathbb{Z}$, split only over $\mathbb{Z}[\frac{1}{2}]$	Written proof; winding and idempotent replay
v7 Prop. 8.13, Rem. 8.14	Gap is the window-mass ratio; same characters with Lebesgue measure do not decorrelate	Exact integrals; finite replay

Nothing in this note is Lean-formalized; the own Lean inventory stays at 33 theorems. The external inputs are the Verdier-duality facts cited above and the Stokes fold classification. Open questions include a quadratic readout on D_+ ; a canonical choice among non-analytic historical readouts, which Theorem 2.6 suggests does not exist without further principles; and a genuinely sign-twisted analytic carrier compatible with the actual Green and Dirac operators. The actual singular Dirac and Pin problems of v12 §8.20–8.24 are untouched here.

References

- [1] P. Scholze, *Six-Functor Formalisms*, lecture notes, arXiv:2510.26269v2 (22 January 2026), Definition 5.1 and Remark 5.4 (p. 32), Example 5.9 (p. 36). <https://arxiv.org/abs/2510.26269>.
- [2] A. Krause, T. Nikolaus and P. Pützstück, *Sheaves on Manifolds*, lecture notes, 7 November 2024, §4.6, Proposition 4.6.9, Remark 4.6.19, Corollary 4.6.20; Theorem 1.1.10.
- [3] M. Kashiwara and P. Schapira, *Sheaves on Manifolds*, Grundlehren 292, Springer, 1990, Chapter III. Standard reference; not retrieved in this round.
- [4] A. Hayami (J.-Y. Huang), *Observation Discriminant and Spherical Fold Image of the Orthogonal-Circle Ruled Surface* (Stokes v5), Zenodo, DOI [10.5281/zenodo.22728902](https://doi.org/10.5281/zenodo.22728902); Theorem 2.4 (“Ordinary folds and the exceptional intersection”) of the registered final PDF.